

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 94-348

WASTE DISCHARGE REQUIREMENTS
FOR
LEMON COVE SANITARY DISTRICT
WASTEWATER TREATMENT AND DISPOSAL FACILITY
TULARE COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Waste Discharge Requirements Order No. 85-173, adopted by the Board on 28 June 1985, prescribes requirements for a discharge of 0.02 million gallons per day of domestic waste from a wastewater treatment facility owned and operated by Lemon Cove Sanitary District (hereafter Discharger).
2. The current wastewater treatment facility consists of a bentonite sealed oxidation pond which is divided into two separate cells. Initial design specifications submitted for the treatment and disposal system show final disposal is accomplished by irrigation of up to 40 acres of adjacent pasture used exclusively for grazing of non-milking animals. However, since construction of the oxidation pond, the flow has not exceeded the evaporation and percolation from the treatment pond and no discharge has occurred to the pasture. Sheep graze within the fenced perimeter of the oxidation pond to control weed growth.
3. Order No. 85-173 is neither adequate to describe this facility nor consistent with current plans and policies of the Board.
4. The thirty-day average daily wastewater flow is less than 10,000 gallons per day. The operational volume of the oxidation pond is about 1.87 million gallons.
5. Average annual precipitation and pan evaporation at this facility are 12 and 91 inches, respectively.
6. The facility and pasture land is in the NE 1/2 of Section 3, T18S, R27E, MDB&M, with surface water drainage to the Kaweah River, as shown on Attachment A, a part of this Order.
7. The site lies within the Kaweah Delta Hydrologic Area (No. 558.10) in the South Valley Floor Hydrologic Unit, as defined on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.

WASTE DISCHARGE REQUIREMENTS
LEMON COVE SANITARY DISTRICT WWTF
TULARE COUNTY

-2-

8. The Board adopted a Water Quality Control Plan for the Tulare Lake Basin (hereafter Basin Plan) which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.
9. The designated beneficial uses of the Kaweah River below Lake Kaweah are agricultural, industrial, municipal and domestic supply; water contact and non-contact water recreation; warm fresh water and wildlife habitat; and groundwater recharge.
10. Depth to first unconfined ground water beneath the site is about eight feet. Source water is from a single community-owned well which is sealed to 50 feet below ground surface and of good quality (EC about 440 μ mhos/cm) except for high nitrate concentrations (NO_3 ranges from 39 to 91 mg/l). The beneficial uses of underlying ground water are municipal and domestic, industrial, and agricultural supply.
11. The action to update waste discharge requirements for this existing facility is exempt from the provisions of the California Environmental Quality Act (CEQA), in accordance with Title 14, California Code of Regulations (CCR), Section 15301.
12. The California Department of Health Services has established statewide reclamation criteria in Title 22, CCR, Section 60301, et seq. (hereafter Title 22) for the use of reclaimed water and has developed guidelines for specific uses.
13. The Board consulted with the Department of Health Services, County Health Department, and Mosquito Abatement District and considered their recommendations regarding public health aspects for use of reclaimed water.
14. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
15. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 85-173 is rescinded and the Lemon Cove Sanitary District, its agents, successors and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

WASTE DISCHARGE REQUIREMENTS
LEMON COVE SANITARY DISTRICT WWTF
TULARE COUNTY

-3-

A. Discharge Prohibitions

1. Discharge of waste to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated domestic wastewater is prohibited.
3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Title 23, CCR, Section 2510, et seq. (hereafter Chapter 15) is prohibited.
4. Grazing of milking animals within the perimeter of the oxidation pond and disposal area is prohibited.

B. Discharge Specifications

1. The monthly average dry weather discharge flow shall not exceed 0.02 million gallons per day.
2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the Discharger's property.
3. As a means of discerning compliance with Discharge Specification B.2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in the oxidation pond shall not be less than 1.0 mg/l.
4. The treatment and disposal facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
5. The effluent from the oxidation pond shall not exceed the following limits:

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Maximum</u>
BOD ₅ ¹	mg/l	70	120
Settleable Solids	ml/l	—	0.5

¹ Five-day, 20° Celsius biochemical oxygen demand.

WASTE DISCHARGE REQUIREMENTS
LEMON COVE SANITARY DISTRICT WWTF
TULARE COUNTY

-4-

6. The maximum electrical conductivity (EC) of the discharge shall not exceed 750 $\mu\text{mhos/cm}$ or the average EC of the source water plus 250 $\mu\text{mhos/cm}$, whichever is less.
7. The effluent in the oxidation pond shall not have a pH less than 6.5 or greater than 8.5.
8. Pond cells shall be managed to prevent breeding of mosquitos. In particular:
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
9. Public contact with wastewater shall be precluded through such means as fences, signs, or other acceptable alternatives.
10. The oxidation pond shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically).
11. On 1 October of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification B.10.
12. Use of reclaimed water shall be limited to flood or furrow irrigation of orchards and vineyards, and irrigation of pasture, fodder, fiber, and seed crops.
13. Public contact with reclaimed water shall be precluded through such means as fences, signs, and irrigation management practices.
14. Reclaimed water for irrigation shall be managed to minimize erosion, runoff, and movement of aerosols from the disposal area.
15. The Discharger may not irrigate effluent during rainfall when the ground is saturated.

WASTE DISCHARGE REQUIREMENTS
LEMON COVE SANITARY DISTRICT WWTF
TULARE COUNTY

-5-

16. Stormwater runoff from the irrigated field shall not be discharged to any surface water drainage course within 30 days of the last application of reclaimed water.
17. Areas irrigated with reclaimed water shall be managed to prevent breeding of mosquitos. More specifically:
 - a. Tail water must be returned and all applied irrigation water must infiltrate completely within a 48-hour period.
 - b. Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.
 - c. Low-pressure and unpressurized pipelines and ditches accessible to mosquitos shall not be used to store reclaimed water.
18. Signs with proper wording of sufficient size shall be placed at areas of access and around the perimeter of all areas used for effluent disposal to alert the public of the use of reclaimed water.
19. There shall be no irrigation or impoundment of reclaimed water within 500 feet of any domestic water well or within 100 feet of any irrigation well unless it is demonstrated to the satisfaction of the Executive Officer that less distance is justified.
20. Application of reclaimed wastewater to the reclamation area shall be at reasonable rates considering the crop, soil, climate, and irrigation management system. The nutrient loading of the reclamation area, including the nutritive value of organic and chemical fertilizers and of the reclaimed water shall not exceed the crop demand.
21. If the Discharger intends to use reclaimed water on crops other than pasture, it shall first submit a written report demonstrating, to the satisfaction of the Executive Officer, that management of reclaimed water and irrigated properties will assure compliance with the terms of this Order.

C. Solids Disposal

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Chapter 15 and approved by the Executive Officer.

WASTE DISCHARGE REQUIREMENTS
LEMON COVE SANITARY DISTRICT WWTF
TULARE COUNTY

-6-

2. Any proposed change in solids use or disposal practice shall be reported to the Executive Officer at least 90 days in advance of the change.

D. Ground Water Limitations

The discharge, in combination with other sources, shall not cause underlying ground water to:

1. Contain waste constituents in concentrations statistically greater than receiving water limits, where specified below, or background water quality where not specified. (For purposes of comparison, background water quality shall be determined when background monitoring provides sufficient data. Quality determined in this manner establishes "water quality protection standards.")
2. Exceed an annual average incremental increase in specific electrical conductivity greater than 3 μ mhos/cm, based on the most recent five-year period, or a maximum of 500 μ mhos/cm, whichever is less.
3. Contain chemicals, heavy metals, or trace elements in concentrations that adversely affect beneficial uses or exceed maximum contaminant levels specified in 22 CCR, Division 4, Chapter 15.
4. Exceed a most probable number of total coliform organisms of 2.2/100 ml over any seven-day period.
5. Exceed concentrations of radionuclides specified in 22 CCR, Division 4, Chapter 15.
6. Contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.

E. Provisions

1. The Discharger shall comply with all the items of the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are part of this Order. This attachment and its individual paragraphs are referred to as "Standard Provision(s)."

WASTE DISCHARGE REQUIREMENTS
LEMON COVE SANITARY DISTRICT WWTF
TULARE COUNTY

-7-

2. The Discharger shall comply with attached Monitoring and Reporting Program No. 94-348, which is part of this Order, and any revisions thereto, as ordered by the Executive Officer.
3. Prior to reclaiming wastewater, the Discharger must first submit a management plan to this office for approval by the Executive Officer a minimum of 120 days prior to irrigating. For properties not included in this Order, a Report of Wastewater Reclamation shall be submitted with the management plan and shall include, but is not limited to, the following:
 - a. Name and address of Landowner or User utilizing reclaimed water. Exact location of land to be irrigated, including maps showing section, township, and range.
 - b. Type of crops, acreage, total quantity of wastewater to be applied (monthly and annually), application method, and seasonal periods of use.
 - c. Nutrient loading rate from wastewater and amount of chemical fertilizers to be used.
4. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the State of incorporation if a corporation, the name and address and telephone number of the persons responsible for contact with the Board, and a statement. The statement shall comply with the signatory paragraph of Standard Provision B.3 and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code. Transfer shall be approved or disapproved in writing by the Executive Officer.

5. The Discharger shall use the best practicable cost-effective control technique currently available to comply with salinity limits specified in this order.
6. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court

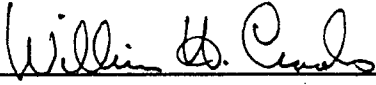
WASTE DISCHARGE REQUIREMENTS
LEMON COVE SANITARY DISTRICT WWTF
TULARE COUNTY

-8-

orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

7. A copy of this Order shall be kept by personnel responsible for operating and maintaining the treatment and disposal facility. Key operating personnel shall be familiar with its contents.
8. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes". Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with 22 CCR, Division 4.
9. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 9 December 1994.



WILLIAM H. CROOKS, Executive Officer

JGW:cjs: 12/9/94

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 94-348

FOR
LEMON COVE SANITARY DISTRICT
WASTEWATER TREATMENT AND DISPOSAL FACILITY
TULARE COUNTY

Specific sample station locations shall be established with concurrence of the Board's staff, and a description of the stations shall be submitted to the Board and attached to this program.

INFLUENT MONITORING

Influent monitoring shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Flow ¹	mgd	Estimated	Weekly
BOD ₅ ²	mg/l	Grab	2/year
Settleable Solids	ml/l	Grab	2/year

¹ Flow measurements shall be taken at the outfall to the oxidation pond.

² Five-day, 20° Celsius biochemical oxygen demand.

EFFLUENT MONITORING

Samples will be collected at the outfall from the oxidation pond and analyzed for the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency¹</u>
Flow	mgd	Estimated	Daily
BOD ₅	mg/l	Grab	Monthly
Settleable Solids	ml/l	Grab	Monthly

¹ Effluent monitoring is required only during those months when treated wastewater is discharged to the disposal area for reclamation.

MONITORING AND REPORTING PROGRAM
LEMON COVE SANITARY DISTRICT WWTF
TULARE COUNTY

-2-

POND MONITORING

The freeboard shall be monitored on the oxidation pond to the nearest tenth of a foot. Monitoring of the pond shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Dissolved Oxygen ¹	mg/l	Grab	Weekly
Freeboard	feet	Observation	Weekly

¹ Samples will be collected from a point opposite of the outfall to the oxidation pond in each cell where water is ponded and analyzed for dissolved oxygen. Samples will be collected between 0800 and 0900 hours.

A permanent marker shall be placed in the pond with calibration indicating the water level at design capacity and available operational freeboard.

In addition, the Discharger shall inspect the condition of the pond once per week and write visual observations in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the pond (e.g., dark sparkling green, dull green, yellow, grey, tan, brown, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the O&M manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

REPORTING

Monitoring results shall be submitted to the Board by the 20th day of the month following sample collection.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner to illustrate clearly whether the discharge complies with waste discharge requirements, including calculation of all averages, etc.

MONITORING AND REPORTING PROGRAM
LEMON COVE SANITARY DISTRICT WWTF
TULARE COUNTY

-3-

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the calculation and reporting of the values required in the discharge monitoring report form. Such increased frequency shall be indicated on the discharge monitoring report form.

By **31 January of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names, titles, certificate grade (if any) and general responsibilities of persons operating and maintaining the wastewater treatment facility.
- b. The names and telephone numbers of persons to contact regarding the facility for emergency and routine situations.
- c. A certified statement of when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who did the calibration (Standard Provision C.4).
- d. A statement certifying whether the current operation and maintenance manual, and contingency plan, reflect the treatment and disposal system as currently constructed and operated, and the dates when these documents were last revised and last reviewed for adequacy.
- e. The total quantity of sludge disposed of during the previous year and ultimate disposal site(s).

The Discharger may also be requested to submit an annual report to the Board with both tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the compliance record. If violations have occurred, the report shall also discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

All reports submitted in response to this Order shall comply with the signatory requirements of Standard Provision B.3.

MONITORING AND REPORTING PROGRAM
LEMON COVE SANITARY DISTRICT WWTF
TULARE COUNTY

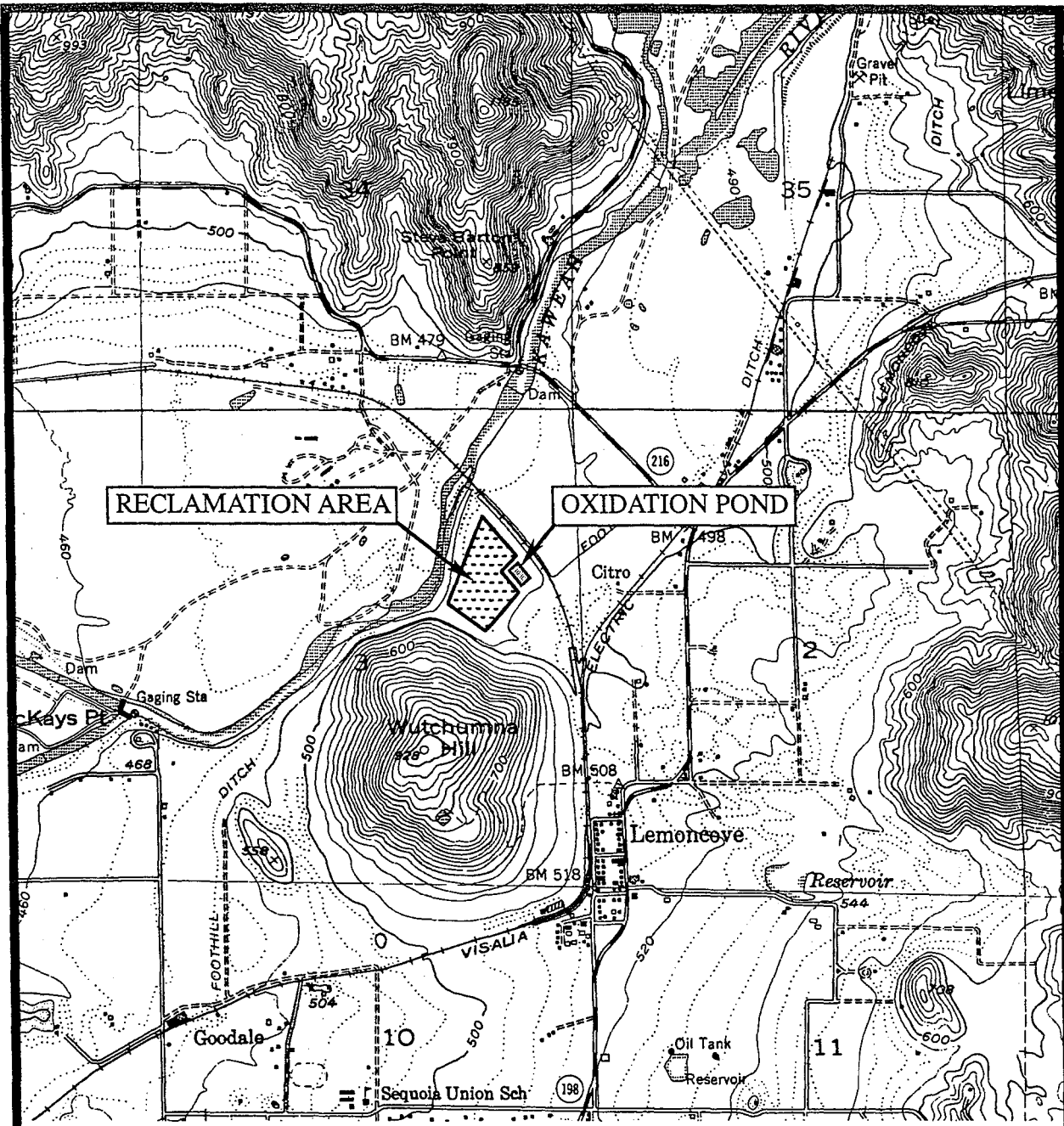
-4-

The Discharger shall implement the above monitoring program on the first day of the month following the effective date of this Order.

Ordered by: William H. Crooks
WILLIAM H. CROOKS, Executive Officer

9 December 1994
(Date)

JGW:cjs: 12/9/94



RECLAMATION AREA

OXIDATION POND



Scale: 1" = 2000'

ATTACHMENT A

Vicinity Map

LEMON COVE SANITARY DISTRICT
WASTEWATER TREATMENT AND DISPOSAL FACILITY
TULARE COUNTY

Section 3, T18S, R27E, MDB&M
Woodlake, 7 1/2' USGS Quad, 1969

INFORMATION SHEET

LEMON COVE SANITARY DISTRICT WASTEWATER TREATMENT AND DISPOSAL FACILITY TULARE COUNTY

The Lemon Cove Sanitary District (District) owns and operates a wastewater treatment and disposal facility about 0.7 miles north of Lemon Cove. Lemon Cove is a small unincorporated community with a population of about 185 persons. The community is in a recreational and agricultural area, surrounded by citrus groves.

The District obtained a Clean Water Grant in 1974 to make improvements to the District's sewage collection and treatment system. The facility was designed to eliminate an open ditch disposal system which passed through town and citrus groves to an irregular shaped pond which at various times overflowed to the Kaweah River. A single 185-foot wide, 300-foot long, 4.5-foot deep Bentonite sealed oxidation pond was constructed and planned disposal was by discharge to about 40 acres of adjacent pasture for non-milking cattle. The oxidation pond was later divided into two cells. The District has not discharged to the pasture since the facility was constructed.

Runoff from the topography flows to the Kaweah River, about 900 feet west of the pond. The designated beneficial uses of the Kaweah River below Lake Kaweah are agricultural, industrial, municipal and domestic supply; water contact and non-contact water recreation; warm fresh water and wildlife habitat; and groundwater recharge.

Depth to ground water is about eight feet below ground surface. However, depth to ground water varies year to year in the area due to its close proximity to the Kaweah River and has been found to be as shallow as six feet below ground surface. Source water is from a single community-owned well which is sealed to 50 feet below ground surface and of good quality (EC about 440 μ mhos/cm) except for high nitrate concentrations (NO_3 ranges from 39 to 91 mg/l). The District is currently investigating methods to treat high nitrate well water or find alternative sources. The beneficial uses of underlying ground water are municipal and domestic, industrial, and agricultural supply.

Soils beneath the facility consist of Tujunga sand. The soils contain considerable gravel and has a high permeability. Average annual precipitation and pan evaporation are about 12 and 91 inches, respectively.

The District grazes sheep to control weeds within the perimeter of its oxidation pond. The sheep are sheared for their wool once per year and have been very effective in controlling weeds in and around the pond. This Order contains discharge specifications to limit nuisances and contains reclamation requirements to prevent health hazards.

This is an existing facility and action to update waste discharge requirements for this facility is exempt from the provisions of the California Environmental Quality Act (CEQA), in accordance with Title 14, California Code of Regulations (CCR), Section 15301.